

BCA-SEMESTER 2ND SYLLABUS

1) Solution of system of linear equations

- a) Direct Method
 - i) Gauss Elimination method
 - ii) Gauss Jordan Method
 - iii) LU Decomposition Method
- b) Indirect (Iterative) Method
 - i) Gauss Jacobi Method
 - ii) Gauss Seidel Method

2) Solution of Algebraic and Transcendental equation

- a) Newton Rapsons Method
- b) Iterative Method
- c) Regula Falsi Method
- d) Bisection (bolzano) Method

3) Interpolation Methods

- a) Lagrange Interpolation method
- b) Newton Forward Interpolation
- c) Newton Backward Interpolation
- d) Central Intermolation

4) Numerical Differentiation

5) Numerical Integration

- a) Trapeqoidal Rule
- b) Simpson's 1/3 Rule
- c) Simpson's 3/8 Rule

6) Solution of ordinary differential equation

- d) Tatlor Series Method
- e) Euler's Method
- f) Modified Euler's Method
- g) Runge – Kutta Method (2nd order)
- h) Runge – Kutta Method (4nd order)

6) Error Analysis

- a) Absalute Error
- b) Relative Error
- c) Percentage Error
- d) Significant Error

NUMERICAL INTEGRATION

➤ Numerical Integration is the process to calculate the value of definite integration

$$\int_a^b f(x) dx$$

- Some method for solving numerical integration.

i) Trapezoidal Rule

ii) Simpson's 1/3 Rule

iii) Simpson's 3/8 Rule

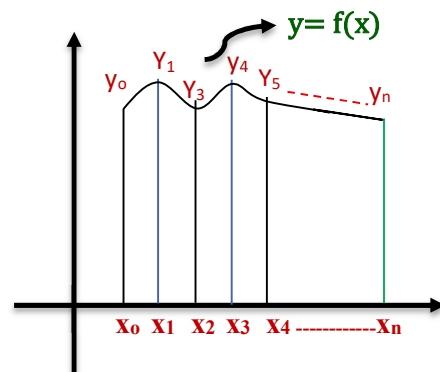
TRAPEZOIDAL RULE

(THEORY)

❖ Explain trapezoidal rule for solving definite integration (Numerical Integration)

➤ Now,

we divide the graph of $f(x)$ into (n) number of small trapezium with height of is trapezium small



$$h = \frac{(b-a)}{n} \text{ where,}$$

n = no of sub interval

$$\int_a^b f(x) dx = \text{Area}(\text{trap}(x_0 y_0 y_1 x_1)) + \text{Area}(\text{trap}(x_1 y_1 y_2 x_2)) + \dots + \text{Area}(\text{trap}(x_{n-1} y_{n-1} y_n x_n))$$

$$\text{➤ } \frac{h}{2}(y_0 + y_1) + \frac{h}{2}(y_1 + y_2) + \frac{h}{2}(y_2 + y_3) + \dots + \frac{h}{2}(y_{n-1} + y_n)$$

$$\text{➤ } \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + y_3 \dots y_{n-1})]$$

(PRACTICAL - 1)

Q1. Find the definite integration by using trapezoidal rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$$

$$h = \frac{1}{4}$$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
F(x)	1	$\frac{16}{4}$	$\frac{4}{5}$	$\frac{16}{25}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{4 \times 2} [(1 + \frac{1}{2}) + 2(\frac{16}{17} + \frac{4}{5} + \frac{16}{25})]$$

$$= \frac{1}{8} [(\frac{3}{2} + \frac{32}{17} + \frac{8}{5} + \frac{32}{25})]$$

$$= \frac{1}{8}(6.2623)$$

$$= 0.7826$$

Answer

(PRACTICAL - 2)

Q1. Find the definite integration by using trapezoidal rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$$

$$h = \frac{1}{4}$$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
F(x)	1	$\frac{16}{4}$	$\frac{4}{5}$	$\frac{16}{25}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{4 \times 2} [(1 + \frac{1}{2}) + 2(\frac{16}{17} + \frac{4}{5} + \frac{16}{25})]$$

$$= \frac{1}{8} [(\frac{3}{2} + \frac{32}{17} + \frac{8}{5} + \frac{32}{25})]$$

$$= \frac{1}{8}(6.2623)$$

$$= 0.7826$$

Answer

(PRACTICAL - 3)

Q1. Find the definite integration by using trapezoidal rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$$

$$h = \frac{1}{4}$$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
F(x)	1	$\frac{16}{4}$	$\frac{4}{5}$	$\frac{16}{25}$	$\frac{1}{2}$
	y ₀	y ₁	y ₂	y ₃	y ₄

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{4 \times 2} [(1 + \frac{1}{2}) + 2(\frac{16}{17} + \frac{4}{5} + \frac{16}{25})]$$

$$= \frac{1}{8} [(\frac{3}{2} + \frac{32}{17} + \frac{8}{5} + \frac{32}{25})]$$

$$= \frac{1}{8}(6.2623)$$

$$= 0.7826$$

Answer

(EXERCISE - 1)

(Objectives)

❖ Trapezoidal Rule – Multiple Choice Questions :-

1. What is the primary use of the Trapezoidal Rule?
A) Differentiation
B) Integration
C) Factorization
D) Interpolation
2. The Trapezoidal Rule approximates the area under a curve by dividing it into:
A) Triangles
B) Rectangles
C) Trapezoids
D) Circles
3. In the Composite Trapezoidal Rule, the step size h is given by:
A) $(a + b)/n$
B) $(b - a)/n$
C) ab/n
D) $(a - b)/n$
4. Which of the following is the formula for the Single Trapezoidal Rule?
A) $(b - a)[f(a) + f(b)]$
B) $\frac{1}{2}(b - a)[f(a) + f(b)]$
C) $(b + a)[f(a) - f(b)]$
D) $\frac{1}{2}(a + b)[f(a) + f(b)]$
5. In the Composite Trapezoidal Rule, the first and last ordinates are counted:
A) Twice
B) Three Times
C) Once
D) Four Times

(EXERCISE - 1)

(Subjectives)

❖ TRAPEZOIDAL RULE – NUMERICAL QUESTIONS :-

Q1. Find the definite integration by using Trapezoidal Rule:

$$\int_0^1 \frac{dx}{1+x^2}$$

Q2. Find the definite integration by using Trapezoidal Rule:

$$\int_0^2 (x^2 + 1) dx$$

Q3. Find the definite integration by using Trapezoidal Rule:

$$\int_0^4 x^2 dx$$

Q4. Find the definite integration by using Trapezoidal Rule:

$$\int_1^3 (x^3 + 2x) dx$$

Q5. Find the definite integration by using Trapezoidal Rule:

$$\int_0^1 e^x dx$$

Q6. Find the definite integration by using Trapezoidal Rule:

$$\int_0^\pi \sin x dx$$

Q7. Find the definite integration by using Trapezoidal Rule:

$$\int_1^2 \frac{dx}{x}$$

SIMPSON'S $\frac{1}{3}$ RULE

(THEORY)

❖ Explain Simpson's $\frac{1}{3}$ rule for solving definite integration (Numerical Integration)

Aage aise hi banana hai yahi formate se